

Exhibit 10: Portfolio Balances across Three Years

Year	1	2	3
Balance from previous year	EUR0	EUR50	EUR1,000
New investment by the investor (cash inflow for the mutual fund) at the start of the year	100	950	0
Net balance at the beginning of year	100	1,000	1,000
Investment return for the year	−50%	35%	27%
Investment gain (loss)	−50	350	270
Withdrawal by the investor (cash outflow for the mutual fund) at the end of the year	0	−350	0
Balance at the end of year	EUR50	EUR1,000	EUR1,270

The **internal rate of return** is the discount rate at which the sum of present values of cash flows will equal zero. In general, the equation may be expressed as follows:

$$\sum_{t=0}^T \frac{CF_t}{(1 + IRR)^t} = 0, \quad (5)$$

where T is the number of periods, CF_t is the cash flow at time t , and IRR is the internal rate of return or the money-weighted rate of return.

A cash flow can be positive or negative; a positive cash flow is an inflow where money flows to the investor, whereas a negative cash flow is an outflow where money flows away from the investor. The cash flows are expressed as follows, where each cash inflow or outflow occurs at the end of each year. Thus, CF_0 refers to the cash flow at the end of Year 0 or beginning of Year 1, and CF_3 refers to the cash flow at end of Year 3 or beginning of Year 4. Because cash flows are being discounted to the present—that is, end of Year 0 or beginning of Year 1—the period of discounting CF_0 is zero.

$$CF_0 = -100$$

$$CF_1 = -950$$

$$CF_2 = +350$$

$$CF_3 = +1,270$$

$$\frac{CF_0}{(1 + IRR)^0} + \frac{CF_1}{(1 + IRR)^1} + \frac{CF_2}{(1 + IRR)^2} + \frac{CF_3}{(1 + IRR)^3} = 0$$

$$= \frac{-100}{1} + \frac{-950}{(1 + IRR)^1} + \frac{+350}{(1 + IRR)^2} + \frac{+1270}{(1 + IRR)^3} = 0$$

$$IRR = 26.11\%$$

The investor's internal rate of return, or the money-weighted rate of return, is 26.11 percent, which tells the investor what she earned on the actual euros invested for the entire period on an annualized basis. This return is much greater than the arithmetic and geometric mean returns because only a small amount was invested when the mutual fund's return was −50 percent.

All the above calculations can be performed using Excel using the =IRR(values) function, which results in an IRR of 26.11 percent.

Money-Weighted Return for a Dividend-Paying Stock

Next, we'll illustrate calculating the money-weighted return for a dividend paying stock. Consider an investment that covers a two-year horizon. At time $t = 0$, an investor buys one share at a price of USD200. At time $t = 1$, he purchases an additional share at a price of USD225. At the end of Year 2, $t = 2$, he sells both shares at a price of USD235. During both years, the stock pays a dividend of USD5 per share. The $t = 1$ dividend is not reinvested. Exhibit 11 outlines the total cash inflows and outflows for the investment.